



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI GOUNDER & CO.

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

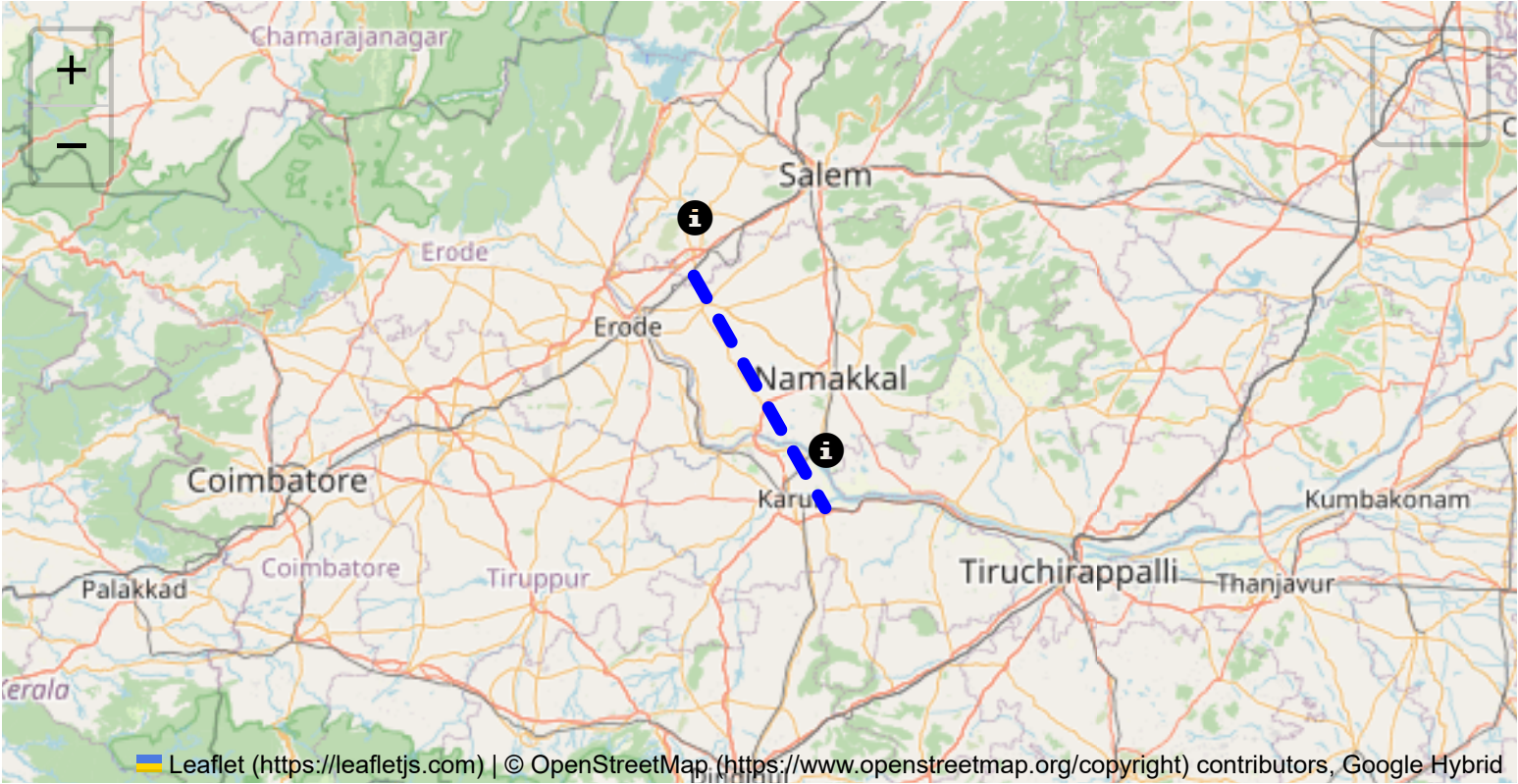
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

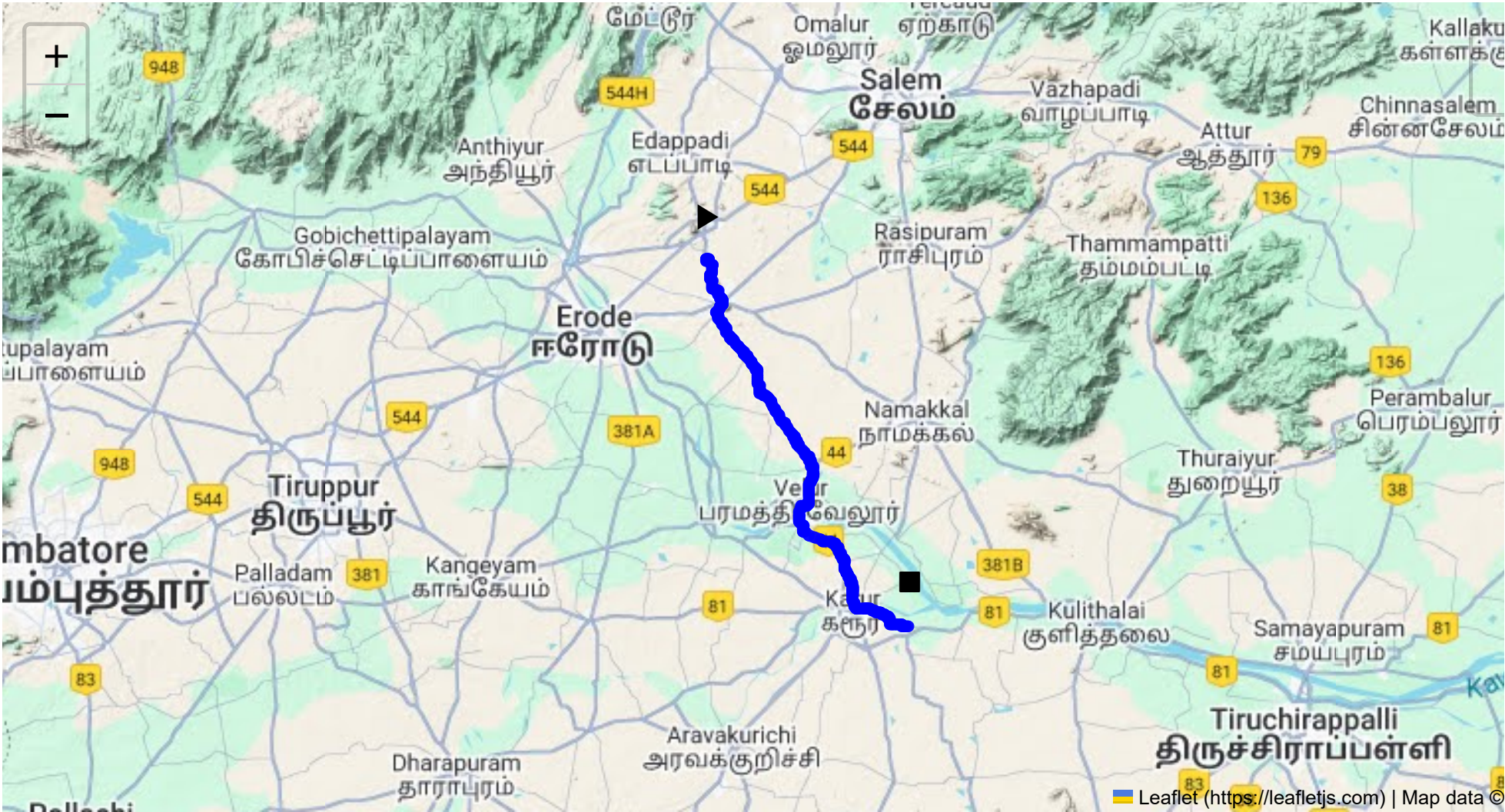
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 73.82 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 2.0 hours
Start: (11.4381, 77.8734)
End: (10.939396, 78.154356)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to W5Q4+P77, Puliur in Tamil Nadu covers approximately 73.82 kilometers, primarily along state and national highways. It passes through rural and semi-urban areas, connecting Sangagiri with Puliur via cities such as Karur. The journey typically involves NH44 for a significant portion and other smaller roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate, with significant rainfall during the monsoon (June to September) and high temperatures from March to May. Potential weather-related hazards include:

- **Heavy rainfalls** causing waterlogging and reduced visibility.
- **Heat waves** leading to road surface degradation and driver fatigue.

3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic congestion generally occurs from 8:00 AM to 10:30 AM and 5:30 PM to 7:30 PM. Urbanized sections, especially near Karur, are more prone to congestion.
- **Congestion-Prone Areas:** Intersections and urban stretches near towns on NH44 and entry points to Karur experience frequent slowdowns.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** NH44 is a major highway with good road conditions, but some connecting roads may have potholes or uneven surfaces, especially after monsoons.
- **Infrastructure:** Facilities such as rest stops and fuel stations are adequate on the highways but limited on smaller roads.

5. Suggestions for Alternative Routes for Emergencies

In case of road closures or severe delays:

- **Alternative Route 1:** Divert via SH71 to avoid NH44 congestion.
- **Alternative Route 2:** Use local roads parallel to NH44 for bypassing major congestion zones, though these might not support heavy vehicles optimally.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits:** Necessary permits required for transporting hazardous materials.
- **Timing Restrictions:** Restrictions may apply during peak urban traffic hours and nighttime in certain zones.
- **Safety Equipment:** Mandatory to have proper signage and safety equipment as per local regulations.

7. Overview of Historical Incidents

- Some historical incidents in the region involve vehicle breakdowns due to mechanical issues and poor road conditions resulting in minor spillage.
- There have been a few recorded traffic accidents near urban centers, mostly due to traffic rule violations or inclement weather.

8. Environmental Considerations and Sensitive Areas

- There are no major ecologically sensitive areas along the route, but care should be taken to avoid pollution.
- Proper waste disposal and maintenance of vehicles to prevent oil spills are crucial for protecting agricultural lands bordering the roads.

9. Analysis of Communication Coverage

- **Coverage:** Mobile network coverage is generally reliable along NH44, though some dead zones may exist on rural sections and secondary roads.
- **Potential Dead Zones:** Parts of connecting roads away from major towns may experience patchy coverage.

10. Estimated Emergency Response Times

- **Urban Areas (e.g., near Karur):** 15-30 minutes given established emergency services.
- **Rural Sections:** Could take up to 45 minutes to 1 hour due to distance from major service areas.

11. Overall Summary of Risk Assessment

This route, while generally safe, does present specific risks associated with weather, road quality, and traffic. Effective pre-trip planning involving route monitoring, awareness of local regulations, and emergency preparedness can mitigate major risks. Regular updates on weather conditions and traffic are advised to ensure a smooth journey, with an emphasis on avoiding peak congestion times. Environmental responsibility and robust communication are also critical to reducing risks associated with transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr
5	Turn	High	11.37862, 77.89488	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16755, 78.01568	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.15579, 78.02169	30 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Blind Spot	Blind Spot	11.14480, 78.02291	10 KM/Hr
11	Turn	High	10.96264, 78.08183	15 KM/Hr
12	Turn	Medium	10.96221, 78.09641	30 KM/Hr
13	Turn	High	10.96229, 78.09658	15 KM/Hr
14	Turn	High	10.96314, 78.09655	15 KM/Hr
15	Turn	Medium	10.96323, 78.09676	30 KM/Hr
16	Turn	High	10.94214, 78.13025	15 KM/Hr
17	Turn	Medium	10.93847, 78.14871	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium
20	hospital	S. K. Hospital, Karur	10.9621165, 78.0840692	30 km/h	Medium
21	hospital	Government Medical College Hospital, Karur	10.9615927, 78.080414	30 km/h	Medium
22	clinic	Madura Nursing Home	10.94078, 78.142504	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
19	school	Cheran Matriculation Higher Secondary School	10.9971306, 78.0743488	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16755, 78.01568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15579, 78.02169



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.96264, 78.08183



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.96221, 78.09641



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.96229, 78.09658



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.96314, 78.09655



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.96323, 78.09676



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.94214, 78.13025



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.93847, 78.14871